Similar questions

Two new questions on the same topic as the exam question, a little more open-ended. They are not past exam questions. Answers are at the end.

1 A rectangle has length $(3+\sqrt{5})$ cm and width $(3-\sqrt{5})$ cm.

(a) Show that the area of the rectangle is 4 cm^2. **[2]**

(b) Find the exact length of a diagonal of the rectangle, giving your answer in the form $a\sqrt{b}$ where b is a prime number. **[3]**

(c) Show that the ratio of the length to the width can be written as $\dfrac{7+3\sqrt{5}}{2}$. **[3]**

2 In this question you must show detailed reasoning.

(a) Show that $(1+\sqrt{2})^2 = 3+2\sqrt{2}$, and write down the value of $\sqrt{3+2\sqrt{2}}$. **[2]**

(b) Express $\dfrac{1}{\sqrt{3+2\sqrt{2}}}$ in the form $a+b\sqrt{2}$, where a and b are integers. **[3]**

(c) Given that $\sqrt{3-2\sqrt{2}} = \sqrt{2}-1$, find the exact value of $\sqrt{3+2\sqrt{2}}+\sqrt{3-2\sqrt{2}}$. **[2]**

Question 1

(a) $(3+\sqrt{5})(3-\sqrt{5}) = 9-5 = 4$

(b) $d^2 = (3+\sqrt{5})^2 + (3-\sqrt{5})^2 = (14+6\sqrt{5}) + (14-6\sqrt{5}) = 28$, so $d = \sqrt{28} = 2\sqrt{7}$

(c) $\dfrac{3+\sqrt{5}}{3-\sqrt{5}} \times \dfrac{3+\sqrt{5}}{3+\sqrt{5}} = \dfrac{14+6\sqrt{5}}{4} = \dfrac{7+3\sqrt{5}}{2}$

Question 2

(a) $(1+\sqrt{2})^2 = 1+2\sqrt{2}+2 = 3+2\sqrt{2}$, so $\sqrt{3+2\sqrt{2}} = 1+\sqrt{2}$

(b) $\dfrac{1}{1+\sqrt{2}} \times \dfrac{\sqrt{2}-1}{\sqrt{2}-1} = \dfrac{\sqrt{2}-1}{2-1} = -1+\sqrt{2}$

(c) $(1+\sqrt{2}) + (\sqrt{2}-1) = 2\sqrt{2}$